

PROGRAM - 09

TITLE: Organize kubernetes pods using labels and namespaces

Project Structure

Program-9

- namespaces.yaml
- pods.yaml

Step 1: Start the minikube

```
1RV24MC018_Ananya_h@user-ThinkPad-E480:~$ minikube start
minikube v1.37.0 on Ubuntu 22.04
Using the docker driver based on existing profile
Starting "minikube" primary control-plane node in "minikube" cluster
Pulling base image v0.0.48 ...
Restarting existing docker container for "minikube" ...

Docker is nearly out of disk space, which may cause deployments to fail! (95% of capacity). You can pass '--force' to skip this check.
Suggestion:

Try one or more of the following to free up space on the device:

1. Run "docker system prune" to remove unused Docker data (optionally with "-a")
2. Increase the storage allocated to Docker for Desktop by clicking on:
   Docker icon > Preferences > Resources > Disk Image Size
3. Run "minikube ssh -- docker system prune" if using the Docker container runtime
Related issue: https://github.com/kubernetes/minikube/issues/9024

Preparing Kubernetes v1.34.0 on Docker 28.4.0 ...
Verifying Kubernetes components...
■ Using image registry.k8s.io/metrics-server/metrics-server:v0.8.0
■ Using image gcr.io/k8s-minikube/storage-provisioner:v5
■ Using image docker.io/kubernetesui/dashboard:v2.7.0
■ Using image docker.io/kubernetesui/metrics-scraper:v1.0.8
💡 Some dashboard features require the metrics-server addon. To enable all features please run:

    minikube addons enable metrics-server

🌟 Enabled addons: storage-provisioner, default-storageclass, metrics-server, dashboard

! /usr/bin/kubectl is version 1.28.15, which may have incompatibilities with Kubernetes 1.34.0.
■ Want kubectl v1.34.0? Try 'minikube kubectl -- get pods -A'
🏁 Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
```

Step 2: Create a namespaces.yaml

```
GNU nano 6.2
apiVersion: v1
kind: Namespace
metadata:
  name: dev
---
apiVersion: v1
kind: Namespace
metadata:
  name: prod
```

Step 3: Create a pods.yaml

```
GNU nano 6.2
apiVersion: v1
kind: Pod
metadata:
  name: frontend-pod
  namespace: dev
  labels:
    app: frontend
    env: dev
spec:
  containers:
    - name: frontend
      image: nginx
      ports:
        - containerPort: 80
---
apiVersion: v1
kind: Pod
metadata:
  name: backend-pod
  namespace: dev
  labels:
    app: backend
    env: dev
spec:
  containers:
    - name: backend
      image: nginx
      ports:
        - containerPort: 80
---
apiVersion: v1
kind: Pod
metadata:
  name: frontend-pod
  namespace: prod
  labels:
    app: frontend
    env: prod
spec:
  containers:
    - name: frontend
      image: nginx
      ports:
        - containerPort: 80
---
apiVersion: v1
kind: Pod
metadata:
  name: backend-pod
  namespace: prod
  labels:
    app: backend
    env: prod
spec:
  containers:
    - name: backend
      image: nginx
      ports:
        - containerPort: 80
```

Step 4: Apply namespaces and pods.yaml

```
1RV24MC018_Ananya_h@user-ThinkPad-E480:~/labprograms/Program-9$ kubectl apply -f namespaces.yaml
namespace/dev created
namespace/prod created
1RV24MC018_Ananya_h@user-ThinkPad-E480:~/labprograms/Program-9$ kubectl apply -f pods.yaml
pod/frontend-pod created
pod/backend-pod created
pod/frontend-pod created
pod/backend-pod created
```

Step 5: Verify pods by the namespace

```
1RV24MC018_Ananya_h@user-ThinkPad-E480:~/labprograms/Program-9$ kubectl get pods -n dev
NAME          READY   STATUS    RESTARTS   AGE
backend-pod    1/1     Running   0           77s
frontend-pod   1/1     Running   0           77s
1RV24MC018_Ananya_h@user-ThinkPad-E480:~/labprograms/Program-9$ kubectl get pods -n prod
NAME          READY   STATUS    RESTARTS   AGE
backend-pod    1/1     Running   0           82s
frontend-pod   1/1     Running   0           82s
```

Step 6: filter pods by the namespaces

```
1RV24MC018_Ananya_h@user-ThinkPad-E480:~/labprograms/Program-9$ kubectl get pods -n dev -l app=frontend
NAME          READY   STATUS    RESTARTS   AGE
frontend-pod   1/1     Running   0           2m24s
1RV24MC018_Ananya_h@user-ThinkPad-E480:~/labprograms/Program-9$ kubectl get pods -n dev -l app=backend
NAME          READY   STATUS    RESTARTS   AGE
backend-pod    1/1     Running   0           2m30s
```

Step 7: Delete the pods

Step 8: Delete all the services

Step 9: Delete all the deployments